Sahil Mohril

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Dedicated and analytical B.Tech Computer Science (Data Science) student with a strong foundation in full-stack backend architecture and engineering scalable data pipelines. Adept at leveraging software engineering principles alongside machine learning frameworks to solve complex, multi-disciplinary problems. Committed to continuous optimization, proactive skill growth, and delivering robust software solutions within collaborative environments.

Education

Vellore Institute of Technology **CGPA: 9.15** Vellore, India
B.Tech in Computer Science and Engineering (Data Science) *2023 – 2027*

Seth M.R. Jaipuria School Score: 93.5% Lucknow, India
Higher Secondary School (ISC) *2022*

Seth M.R. Jaipuria School Score: 96.0% Lucknow, India
Secondary School (ICSE) *2020*

Technical Skills

Languages: Java, Python, C, C++, SQL
Web & Backend: Spring Boot, React, HTML, REST APIs
Databases: MySQL, PostgreSQL, MongoDB
Data Science & ML: Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Exploratory Data Analysis
Big Data & Cloud: Apache Spark (PySpark), Apache Kafka, AWS
Developer Tools: Git, GitHub, Databricks, Tableau, VS Code

Projects

Predictive Maintenance System for Rotating Machinery | *Python, Scikit-learn, Signal Processing* **[GitHub]**

- Built an end-to-end predictive maintenance pipeline combining vibration signal processing with machine learning to estimate Remaining Useful Life (RUL) of bearings.
- Engineered time-domain and frequency-domain features from raw sensor signals and trained regression models to predict degradation trends.
- Validated pipeline on benchmark bearing run-to-failure datasets, improving early fault-detection accuracy over baseline threshold methods.

MapReduce Engine in Java | *Java, Multithreading, Distributed Systems* **[GitHub]**

- Designed and implemented a distributed MapReduce framework from scratch in Java, supporting custom Mapper and Reducer interfaces similar to Hadoop's programming model.
- Built a master-worker architecture to partition input data, schedule map and reduce tasks across worker threads/nodes, and shuffle intermediate key-value pairs.
- Implemented fault tolerance with task re-execution on worker failure and used the engine to run word-count and large-scale aggregation, demonstrating linear speedup with added workers.

CabConnect – Cab Booking Application | *Spring Boot, Java, MySQL* **[GitHub]**

- Built a full-stack cab booking management platform using Spring Boot, featuring ride booking, user management, and real-time location tracking.
- Designed and exposed RESTful APIs for booking, driver assignment, and trip status updates, integrated with a persistent MySQL backend.
- Implemented role-based access for riders and drivers and structured the backend using layered architecture (Controller-Service-Repository) for maintainability.

Certifications

Oracle Certified Java Professional (SE 11) | *Oracle*

Data Science Course, Cognizance '24 | *IIT Roorkee*

NumPy and Pandas Masterclass | *Udemy*

MySQL Bootcamp | *Udemy*